

Target dossier

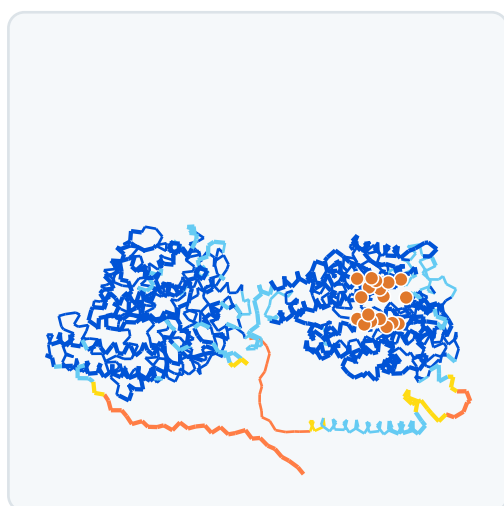
ACE: Angiotensin-converting enzyme

UniProt P12821 · 1306 residues · ChEMBL CHEMBL1808 · linked findings: Blood pressure and vascular risk, Diabetic kidney disease risk

Key insights

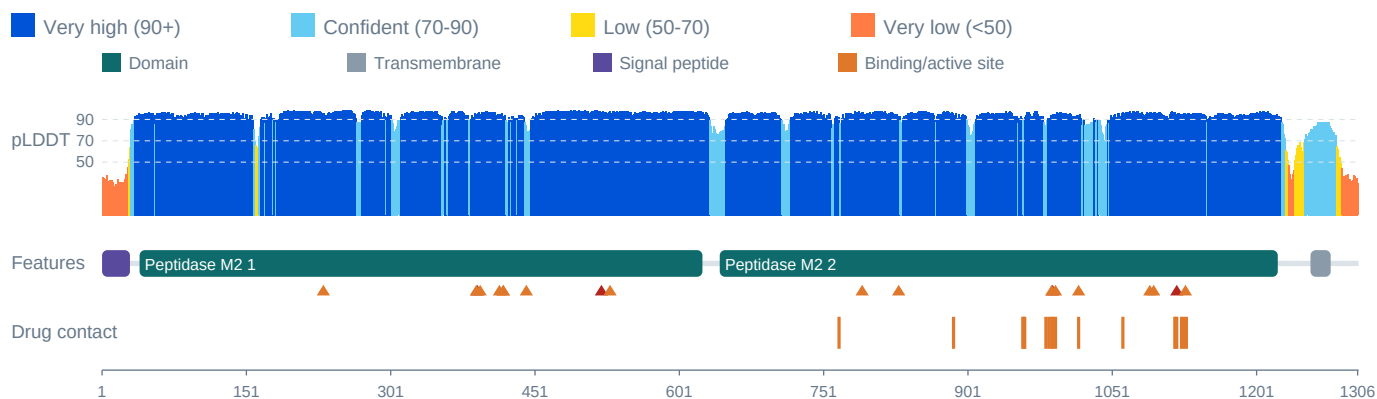
- **Druggable site mapped.** 18 residues lie within 4.5 Å of the ligand LPR in PDB 1O86; AlphaFold confidence there is 93 (high: the AlphaFold model reproduces the site well enough for structure-based screening and docking).
- **Flexible regions** (pLDDT below 50) at residues 1-27, 1289-1306: likely disordered or unstructured; exclude them from pocket searches and expect them to be absent in crystal structures.
- **Validated and crowded mechanism.** 10 approved drugs (small molecule 10); a new entrant needs differentiation in selectivity, dosing interval, delivery or a defined patient subgroup.
- **Patient demand signal.** 18 of 21 screened patients (86%) carry a finding linked to this target: a recruitment and market signal detected non-invasively.
- **Clinical activity.** 8 recruiting trial(s) near the clinic for the linked findings.

Structure



Mean pLDDT	90.9
Confident residues (pLDDT 70+)	94%
Disordered regions	1-27, 1289-1306
Drug-contact residues	18 (PDB 1O86)
Contact-site pLDDT	93.1

AlphaFold model projected on its two principal axes, coloured by confidence; orange dots mark residues touching the drug in the experimental complex.



Drug-contact residues (UniProt numbering): E767, Q886, H958, A959, S960, D982, V985, H988, E989, H992, E1016, F1062, K1116, F1117, H1118, V1123, Y1125, Y1128

Function (UniProt)

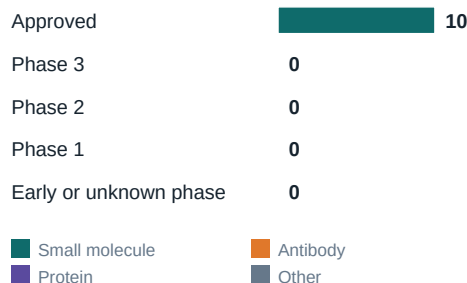
Dipeptidyl carboxypeptidase that removes dipeptides from the C-terminus of a variety of circulating hormones, such as angiotensin I, bradykinin or enkephalins, thereby playing a key role in the regulation of blood pressure, electrolyte homeostasis or synaptic plasticity.

Composed of two similar catalytic domains, each possessing a functional active site, with different selectivity for substrates.

Why the eye points here

Converts angiotensin I to angiotensin II; ACE inhibitors are first-line for hypertension with albuminuria in diabetes. Hypertension and kidney signals.

Drug landscape



Drug	Stage	Action	Modality	ChEMBL
benazepril	Approved	inhibitor	small molecule	CHEMBL1694
captopril	Approved	inhibitor	small molecule	CHEMBL1560
cilazapril	Approved	inhibitor	small molecule	CHEMBL515606
enalapril	Approved	inhibitor	small molecule	CHEMBL1200659
enalaprilat	Approved	inhibitor	small molecule	CHEMBL3989406
fosinopril	Approved	inhibitor	small molecule	CHEMBL3039596
imidapril	Approved	inhibitor	small molecule	CHEMBL317094
lisinopril	Approved	inhibitor	small molecule	CHEMBL419213
moexipril	Approved	inhibitor	small molecule	CHEMBL1200534
perindopril	Approved	inhibitor	small molecule	CHEMBL3544986

In the Insight Rx therapy guide

- **ACE inhibitor or ARB** (Guideline first-line): Recommended for hypertension in diabetes, and first choice when albuminuria is present. Use one, never both. ADA Standards of Care, Sections 10 and 11; KDIGO 2022 Diabetes in CKD

Recruiting clinical trials

ClinicalTrials.gov, searched by finding near the clinic; no patient data is sent. The site confirms full eligibility.

Finding	Trial	Phase / sponsor	Sites
Blood pressure and vascular risk	Implementing AI-based Glaucoma Screening Within Federally Qualified Health Centers NCT06629649 Department of Ophthalmology and Visual Sciences, University of Alabama at Birmingham, Birmingham, Alabama	NA · University of Alabama at Birmingham	1 site(s)
Blood pressure and vascular risk	Improving the Collaborative Health of Adult Dyads NCT05370014 University of South Carolina, Columbia, South Carolina	NA · University of South Carolina	1 site(s)
Blood pressure and vascular risk	EASi-PROTKT™ - A Study to Test Vicadrost (BI 690517) Taken Together With Empagliflozin in People With Type 2 Diabetes, High Blood Pressure, and Cardiovascular Disease NCT07064473 Pinnacle Research Group, LLC, Anniston, Alabama	Phase 3 · Boehringer Ingelheim	1143 site(s)

Finding	Trial	Phase / sponsor	Sites
Blood pressure and vascular risk	Cardiometabolic Disease and Substrate Metabolism NCT07565727 University of Tennessee Graduate School of Medicine, Knoxville, Tennessee	N/A · University of Tennessee Graduate School of Medicine	1 site(s)
Diabetic kidney disease risk	Sotagliflozin to Slow Kidney Function Decline in Persons With Type 1 Diabetes and Diabetic Kidney Disease NCT06217302 Stanford University Medical Center, Stanford, California	Phase 3 · Alessandro Doria	19 site(s)
Diabetic kidney disease risk	PREvention of Cardiovascular and Diabetic kidney Disease in Type 2 Diabetes NCT05390892 Longwood Research LLC, Huntsville, Alabama	Phase 4 · Brigham and Women's Hospital	36 site(s)
Diabetic kidney disease risk	Study to Assess the Effects of ALN-ANG3 in Adult Participants With Diabetic Kidney Disease NCT07271186 Apogee Clinical Research, LLC, Huntsville, Alabama	Phase 2 · Regeneron Pharmaceuticals	58 site(s)
Diabetic kidney disease risk	Safety and Tolerability of Vertebral Bone Marrow-derived Mesenchymal Stem Cells (BM-MSK) in Real World Scenarios of Patients With Chronic Kidney Disease (CKD) NCT06752577 Mayo Clinic, Jacksonville, Florida	N/A · Mayo Clinic	1 site(s)

Sources

UniProt; AlphaFold Protein Structure Database (DeepMind/EMBL-EBI, CC-BY 4.0); RCSB PDB; ChEMBL mechanisms (EMBL-EBI); ClinicalTrials.gov. Contacts computed at 4.5 Å between heavy atoms of the drug and the target, mapped to UniProt numbering by sequence alignment.